

Role of dietary lignans in the reduction of breast cancer risk.



Lignans are a large group of fiber-associated phenolic compounds widely distributed in edible plants. Some of the ingested plant lignans are converted by intestinal microbiota to enterolignans, enterodiols (END) and enterolactone (ENL), the latter of which has been thought to be the major biologically active lignan, and suggested to be associated with low risk of breast cancer. In line with this, administration of plant lignans which are further metabolized to ENL, or ENL as such, have been shown to inhibit or delay the growth of experimental mammary cancer. The mechanism of anticarcinogenic action of ENL is not yet fully understood, but there is intriguing evidence for ENL as a modulator of estrogen signaling. These findings have generated interest in the use of lignans as components of breast cancer risk reducing functional foods. Identification of target groups, who would benefit most, is of pivotal importance. Therefore, further identification and validation of relevant biomarkers, which can be used as indicators of lignan or ENL action and breast cancer risk reduction at different stages of the disease, are of importance.